

COAL INDIA LIMITED (CIL) - COMPREHENSIVE ANNUAL REVIEW

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Section 1: Macro Production, Offtake and National Grid Supply

Coal India Limited (CIL) registered record raw coal production of 703.20 Million Tonnes (MT) during FY 2023-24, achieving double-digit growth of 10.1% over 638.60 MT in FY 2022-23. Offtake attained 753.50 MT, ensuring peak thermal inventory across all national generating stations. Opencast mining produced 665.40 MT (94.62%), while underground operations contributed 37.80 MT (5.38%). Mechanized longwall and continuous miners at ECL and SECL boosted underground productivity.

Subsidiary	OC (MT)	UG (MT)	Total (MT)	Target (MT)	Growth %	Primary Basin
Mahanadi Coalfields (MCL)	181.50	11.80	193.30	190.00	+11.9%	Talcher / Ib Valley
South Eastern Coalfields (SECL)	155.80	11.20	167.00	170.00	+13.2%	Korba / Mand-Raigarh
Northern Coalfields (NCL)	131.00	0.00	131.00	131.00	+6.8%	Singrauli / Moher Basin
Central Coalfields (CCL)	82.90	1.10	84.00	84.00	+14.2%	North & South Karanpura
Western Coalfields (WCL)	57.10	3.20	60.30	62.00	+4.5%	Wardha / Umrer Basin
Bharat Coking Coal (BCCL)	39.80	1.30	41.10	41.00	+17.4%	Jharia Coalfield
Eastern Coalfields (ECL)	25.90	9.20	35.10	37.00	+4.8%	Raniganj / Rajmahal
North Eastern Coalfields (NEC)	0.50	0.00	0.50	0.60	-2.1%	Makum Coalfield, Assam
CIL Total Aggregate	665.40	37.80	703.20	715.60	+10.1%	All Gondwana Basins

Section 2: Overburden Removal (OBR) and Stripping Ratio Dynamics

Composite Overburden Removal (OBR) achieved a historic 1,650.40 Million Cubic Metres (M.Cum) during FY 2023-24, representing an expansion of 21.6% compared to 1,357.20 M.Cum logged in FY 2022-23. The company-wide average stripping ratio stabilized at 2.48 m³/t. WCL logged the highest stripping ratio of 4.21 m³/t due to deep pitching seams, whereas MCL sustained the lowest stripping ratio of 1.18 m³/t across its Talcher mega-pits. Dragline deployment in NCL Singrauli and SECL Korba achieved an availability index of 84.6%.

Section 3: Coal Beneficiation & Gross Calorific Value (GCV) Banding

Gross Calorific Value (GCV) grade distribution during FY 2023-24 showed that 62.4% of total dispatches fell within the G10 to G14 grades (GCV 3101 - 4600 kcal/kg). High-grade non-coking coal (G1-G6, >5500 kcal/kg) from Raniganj and Korba was earmarked for chemical synthesis and fertilizer complexes. Madhuband washery (5.0 MTY) and Dahibari washery (1.6 MTY) beneficiated coking coals to deliver 18.0% ash coking feed to steel plants.

Section 4: Coal Bed Methane (CBM) & Underground Gasification

Commercial CBM development progressed in Jharia Block-I and Raniganj South. Hydro-fracturing stimulation in deep Barakar Seams (Seam XVIII to Seam XV) demonstrated desorption gas flows averaging 3,800 to 7,200 m³/day per well. Estimated Gas-In-Place (GIP) across CIL leaseholds stands at 25.40 Billion Cubic Metres (BCM).

Section 5: Environmental Stewardship & First Mile Connectivity (FMC)

Under the First Mile Connectivity (FMC) initiative, CIL commissioned 35 rapid loading silos integrated with rail lines, eliminating 45,000 truck movements per day. Satellite remote sensing conducted by CMPDI across 76 major opencast projects revealed that 1,850 hectares of biologically reclaimed land achieved dense vegetation canopy, reducing ambient PM10 by 24%.